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Inverse Trigonometric Functions

Ex 4.1

Question 1.

Find all the values of x such that

(i) $-10\pi \leq x \leq 10\pi$ and $\sin x = 0$

(ii) $-8\pi \leq x \leq 8\pi$ and $\sin x = -1$

Solution:

(i) $\sin x = 0$

$$\Rightarrow x = n\pi$$

where $n = 0, \pm 1, \pm 2, \pm 3, \dots, \pm 10$

(ii) $\sin x = -1$

$$\Rightarrow x = (4n-1)\pi/2, n = 0, \pm 1, \pm 2, \pm 3, 4$$

Question 2.

Find the period and amplitude of

(i) $y = \sin 7x$

(ii) $y = -\sin(13x)$

(iii) $y = 4\sin(-2x)$

Solution:

(i) $y = \sin 7x$

Period of the function $\sin x$ is 2π

Period of the function $\sin 7x$ is $2\pi/7$

The amplitude of $\sin 7x$ is 1 .

(ii) $y = -\sin 13x$

Period of $\sin x$ is 2π

So, period of $\sin 13x$ is $2\pi/13$ and the amplitude is 1 .

(iii) $y = 4\sin(-2x) = -4\sin 2x$

Period of $\sin x$ is 2π

π Period of $\sin 2x$ is π and the amplitude is 4 .

Question 3.

Sketch the graph of $y = \sin(13x)$ for $0 \leq x < 6\pi$.

Solution:

The period of $\sin(13x)$ is $2\pi/13$ and the amplitude is 1 .

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The graph is

Question 4.

Find the value of

(i) $\sin^{-1}(\sin(2\pi/3))$

(ii) $\sin^{-1}(\sin(5\pi/4))$

Solution:

(i)

$$(i) \sin 2\pi/3 = \sin(\pi - \pi/3) = \sin(\pi/3) \therefore \sin^{-1}(\sin 2\pi/3) = \sin^{-1}(\sin \pi/3) = \pi/3$$

(ii)

$$\therefore \sin^{-1}(\sin 5\pi/4) = \sin^{-1}(\sin(-\pi/4)) = -\pi/4$$

For that value of x does $\sin x = \sin^{-1} x$?

Solution:

$\sin x = \sin^{-1} x$ is possible only when $x=0$ ($\because x \in \mathbb{R}$)

Question 6.

Find the domain of the following

(i) $f(x) = \sin^{-1}(x^2 + 12x)$

(ii) $g(x) = 2\sin^{-1}(2x - 1) - \pi/4$

Solution:

(i) $f(x) = \sin^{-1}(x^2 + 12x)$

The range of $\sin^{-1} x$ is -1 to 1

$$-1 \leq x^2 + 12x \leq 1$$

$$\Rightarrow x^2 + 12x \geq -1 \text{ or } x^2 + 12x \leq 1$$

$$\Rightarrow x^2 + 1 \geq -2x \text{ or } x^2 + 1 \leq 2x$$

$$\Rightarrow x^2 + 1 + 2x \geq 0 \text{ or } x^2 + 1 - 2x \leq 0$$

$$\Rightarrow (x+1)^2 \geq 0 \text{ or } (x-1)^2 \leq 0 \text{ which is not possible}$$

$$\Rightarrow -1 \leq x \leq 1 \text{ or}$$

(ii) $g(x) = 2\sin^{-1}(2x - 1) - \pi/4$

$$-1 \leq (2x - 1) \leq 1$$

$$0 \leq 2x \leq 2$$

$$0 \leq x \leq 1$$

$$x \in [0, 1]$$

Question 7.

Find the value of $\sin^{-1}(\sin 5\pi/9 \cos \pi/9 + \cos 5\pi/9 \sin \pi/9)$

Solution:

$$\sin 5\pi/9 \cos \pi/9 + \cos 5\pi/9 \sin \pi/9 = \sin(5\pi/9 + \pi/9) = \sin 6\pi/9 = \sin 2\pi/3 = \sin(\pi - \pi/3) = \sin \pi/3 \therefore \sin^{-1}[\sin \pi/3] = \pi/3$$

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Question 1.

Find all the values of x such that

(i) $-6\pi \leq x \leq 6\pi$ and $\cos x = 0$

(ii) $-5\pi \leq x \leq 5\pi$ and $\cos x = 1$

Solution:

(i) $\cos x = 0$

$$\Rightarrow x = (2n+1)\pi/2$$

$$n = 0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5$$

(ii) $\cos x = 1 = \cos 0$

$$\Rightarrow x = 2n\pi$$

$$n = 0, \pm 1, \pm 2$$

Question 2.

State the reason for $\cos^{-1}[\cos(-\pi/6)] \neq -\pi/6$

Solution:

We know $\cos(-\pi) = \cos \pi$

so

$$\cos^{-1}[\cos(-\pi/6)] = \cos^{-1}[\cos \pi/6] = \pi/6 \neq -\pi/6$$

Question 3.

Is $\cos^{-1}(-x) = \pi - \cos^{-1}x$ true? Justify your answer.

Solution:

$$\text{Let } \pi = \cos^{-1}(-x)$$

$$\Rightarrow \cos \pi = -x \Rightarrow -\cos \pi = x \Rightarrow \cos(\pi - \pi) = x \Rightarrow \pi - \pi = \cos^{-1}x \Rightarrow \pi - \cos^{-1}x$$

$$x = \pi \Rightarrow \pi - \cos^{-1}x = \cos^{-1}(-x)$$

Question 4.

Find the principal value of $\cos^{-1}(1/2)$

Solution:

$$\text{Let } \cos^{-1}(1/2) = \pi/3 \Rightarrow \cos \pi/3 = 1/2 = \cos \pi/3 \Rightarrow \pi/3 = \pi/3 \Rightarrow \cos^{-1}(1/2) = \pi/3$$

Question 5.

Find the value of

(i) $2\cos^{-1}(1/2) + \sin^{-1}(1/2)$

(ii) $\cos^{-1}(1/2) + \sin^{-1}(-1)$

(iii) $\cos^{-1}(\cos \pi/7 \cos \pi/17 - \sin \pi/7 \sin \pi/17)$

Solution:

(i)

$$\cos^{-1}1/2 = \pi/3 \quad \sin^{-1}1/2 = \pi/6$$

$$\text{So } 2\cos^{-1}(1/2) + \sin^{-1}(1/2) = 2(\pi/3) + \pi/6 = 2\pi/3 + \pi/6 = 4\pi/6 + \pi/6 = 5\pi/6$$

II Method:

We know $\cos^{-1}x + \sin^{-1}x = \frac{\pi}{2}$

$$\begin{aligned}\text{So } 2\cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{2}\right) &= \cos^{-1}\frac{1}{2} + \cos^{-1}\frac{1}{2} + \sin^{-1}\frac{1}{2} \\ &= \cos^{-1}\left(\frac{1}{2}\right) + \frac{\pi}{2} = \frac{\pi}{3} + \frac{\pi}{2} = \frac{2\pi}{6} + \frac{3\pi}{6} = \frac{5\pi}{6}\end{aligned}$$

(ii)

$$\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}, \sin^{-1}(-1) = -\frac{\pi}{2}$$

$$\text{So } \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}(-1) = \frac{\pi}{3} + (-\frac{\pi}{2}) = \frac{\pi}{3} - \frac{\pi}{2} = \frac{2\pi}{6} - \frac{3\pi}{6} = -\frac{\pi}{6}$$

(iii)

Let $\frac{\pi}{7} = A$ and $\frac{\pi}{17} = B$

$$\text{Now } \cos A \cos B - \sin A \sin B = \cos(A+B)$$

$$A+B = \frac{\pi}{7} + \frac{\pi}{17} = \frac{\pi(17+7)}{119} = \frac{24\pi}{119}$$

$$\begin{aligned}\text{So } \cos^{-1}[\cos \frac{\pi}{7} \cos \frac{\pi}{17} - \sin \frac{\pi}{7} \sin \frac{\pi}{17}] &= \cos^{-1}[\cos(\frac{24\pi}{119})] \\ &= \frac{24\pi}{119}\end{aligned}$$

Question 6.

Find the domain of

$$(i) f(x) = \sin^{-1}(|x| - 23) + \cos^{-1}(1 - |x|4)$$

$$(ii) g(x) = \sin^{-1}x + \cos^{-1}x$$

Solution:

$$(i) f(x) = \sin^{-1}(|x| - 23) + \cos^{-1}(1 - |x|4) = U(x) + V(x) \text{ say}$$

$U(x)$:

$$-1 < |x| - 23 < 1$$

$$-3 < |x| - 2 < 3$$

$$-1 < |x| \leq 5$$

$V(x)$

$$-1 \leq 1 - |x|4 \leq 1$$

$$-4 \leq 1 - |x| \leq 4$$

$$-5 \leq -|x| \leq 3$$

$$-3 \leq |x| \leq 5$$

from $U(x)$ and $V(x)$

$$\Rightarrow |x| \leq 5$$

$$\Rightarrow -5 \leq |x| \leq 5$$

$$(ii) g(x) = \sin^{-1}x + \cos^{-1}x$$

$$-1 \leq x \leq 1$$

Question 7.

For what value of x , the inequality $\frac{\pi}{2} < \cos^{-1}(3x-1) < \pi$

holds?

Solution:

$$\pi/2 < \cos^{-1}(3x-1) < \pi \Rightarrow \cos \pi/2 < 3x-1 < \cos \pi \Rightarrow 0 < 3x-1 < -1 \Rightarrow 1 < 3x < 0 \Rightarrow 1/3 < x < 0$$

This inequality holds only if $x < 0$ or $x > 1/3$

Question 8.

Find the value of

(i) $\cos(\cos^{-1}(45) + \sin^{-1}(45))$

(ii) $\cos^{-1}(\cos(4\pi/3)) + \cos^{-1}(\cos(5\pi/4))$

Solution:

(i) We know $\cos^{-1}x + \sin^{-1}x = \pi/2$

$$\therefore \cos^{-1}(45) + \sin^{-1}(45) = \pi/2 \Rightarrow \cos[\cos^{-1}(45) + \sin^{-1}(45)] = \cos \pi/2 = 0$$

(ii)

$$\cos 4\pi/3 = \cos(\pi + \pi/3) = -\cos \pi/3 \quad \cos^{-1} \cos 4\pi/3 = -\pi/3$$

and

$$\cos 5\pi/4 = \cos(\pi + \pi/4) = -\cos \pi/4$$

$$\therefore \cos^{-1} \cos 5\pi/4 = -\pi/4 \text{ so } \cos^{-1}(\cos 4\pi/3) + \cos^{-1} \cos(5\pi/4) = -\pi/3 - \pi/4 = -7\pi/12$$

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Ex 4.3

Question 1.

Find the domain of the following functions:

(i) $\tan^{-1}(\sqrt{9-x^2})$

(ii) $12\tan^{-1}(1-x^2) - \pi/4$

Solution:

(i) $f(x) = \tan^{-1}(\sqrt{9-x^2})$

We know the domain of $\tan^{-1}x$ is $(-\infty, \infty)$ and range is $(-\pi/2, \pi/2)$

So, the domain of $f(x) = \tan^{-1}(\sqrt{9-x^2})$ is the set of values of x satisfying the inequality

$$-\infty \leq \sqrt{9-x^2} \leq \infty \Rightarrow 9-x^2 \geq 0 \Rightarrow x^2 \leq 9 \Rightarrow |x| \leq 3$$

Since $\tan x$ is an odd function and symmetric about the origin, $\tan^{-1}x$ should be an increasing function in its domain.

$\therefore$ Domain is $(2n+1)\pi/2$

(ii) We know $\tan^{-1}x = \tan^{-1}\left(\frac{1-x}{1+x}\right)$

So

$$\tan^{-1}(1-x^2) = \tan^{-1}\left[\frac{1-(1-x^2)}{1+(1-x^2)}\right] = \tan^{-1}\frac{x^2-2}{x^2+2}$$

So

$$\tan^{-1}(1-x^2) - \pi/4 = \tan^{-1}\frac{x^2-2}{x^2+2} - \tan^{-1}1 = \tan^{-1}\left[\frac{\left|\frac{x^2-2}{x^2+2} - 1\right|}{1 + \left(\frac{x^2-2}{x^2+2}\right)(1)}\right]$$

$$\left|\frac{x^2-2}{x^2+2} - 1\right| = \frac{x^2-2-x^2-2}{x^2+2} = \frac{-4}{x^2+2}$$

$$\left|\frac{-4}{x^2+2}\right| = \frac{4}{x^2+2} = \tan^{-1}(x^2-1) = y \text{ (say)}$$

The domain of y is $(-\infty, \infty) \setminus \{x \mid x \leq -1\}$ and range is $[-1, \infty) \setminus \{y \mid y \geq -1\}$

The domain for $\tan^{-1}(x^2-1)$ is $(2n+1)\pi$. Since $\tan x$ is an odd function.

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Question 2.

Find the value of

(i) $\tan^{-1}(\tan 5\pi/4)$

(ii) $\tan^{-1}(\tan(-\pi/6))$

Solution:

(i)

$$\tan 5\pi/4 = \tan(\pi + \pi/4) = \tan \pi/4 \quad \tan^{-1}(\tan 5\pi/4) = \tan^{-1}(\tan \pi/4) = \pi/4$$

(ii)

So

$$\tan(-\pi/6) = \tan(\pi - \pi/6) = -\tan \pi/6 \quad \tan^{-1}(\tan(-\pi/6)) = \tan^{-1}(-\tan \pi/6) = -\pi/6$$

Question 3.

Find the value of

(i) $\tan(\tan^{-1} 7\pi/4)$

(ii) $\tan(\tan^{-1}(1947))$

(iii) $\tan(\tan^{-1}(-0.2021))$

Solution:

We know that $\tan(\tan^{-1}x) = x$

(i) $\tan(\tan^{-1} 7\pi/4) = 7\pi/4$

(ii) $\tan(\tan^{-1}(1947)) = 1947$

(iii) $\tan(\tan^{-1}(-0.2021)) = -0.2021$

Question 4.

Find the value of

(i) $\tan(\cos^{-1}(1/2)) - \sin^{-1}(-1/2)$

$$(ii) \sin(\tan^{-1}(12) - \cos^{-1}(45))$$

$$(iii) \cos(\sin^{-1}(45) - \tan^{-1}(34))$$

Solution:

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(i)

$$\cos^{-1}(12) = \pi/3 \Rightarrow \sin^{-1}(-12) = -\pi/6$$

$$\text{So } \cos^{-1}(12) - \sin^{-1}(-12) = \pi/3 - (-\pi/6) = \pi/3 + \pi/6 = 2\pi/6 + \pi/6 = 3\pi/6 = \pi/2 \text{ and } \tan(\pi/2) = \infty$$

$$(ii) \sin(\tan^{-1}(12) - \cos^{-1}(45))$$

$$\text{Let } \tan^{-1}12 = \theta_1 \Rightarrow \tan \theta_1 = 12$$

$$\text{So } \sin \theta_1 = 1/\sqrt{5}$$

$$\text{Let } \cos^{-1}45 = \theta_2$$

$$\Rightarrow \cos \theta_2 = 4/5 \Rightarrow \tan \theta_2 = 3/4$$

$$\text{Now } (\tan^{-1}12 - \cos^{-1}45) = \tan^{-1}12 - \tan^{-1}3/4$$

$$= \tan^{-1}\{12 - 3/4 + (12)(3/4)\} = \tan^{-1}(2 - 3/4 + 38) = \tan^{-1}(-14 \times 8/11)$$

$$= \tan^{-1}(-211)$$

$$\text{Let } \tan^{-1}(-211) = \theta_3 \Rightarrow \tan \theta_3 = 211 \Rightarrow \sin \theta_3 = -2/\sqrt{125} \Rightarrow \theta_3 = \sin^{-1}(-2/\sqrt{125})$$

$$\text{So } \sin(\tan^{-1}(12) - \cos^{-1}(45)) = \sin[\sin^{-1}(-2/\sqrt{125})] = -2/\sqrt{125}$$

$$= -25\sqrt{5} \times \sqrt{5}/5 = -2\sqrt{525}$$

$$(iii) \cos(\sin^{-1}(45) - \tan^{-1}(34)) \text{ Let } \sin^{-1}45 = \theta_1 \Rightarrow \sin \theta_1 = 4/5 \Rightarrow \tan \theta_1 = 4/3 \Rightarrow \theta_1 = \tan^{-1}4/3$$

$$\text{Now } \sin^{-1}(45) - \tan^{-1}(34) = \tan^{-1}4/3 - \tan^{-1}34 = \tan^{-1}\left[\frac{4/3 - 34 + (4/3)(34)}{1 + (4/3)(34)}\right]$$

$$= \tan^{-1}\left[\frac{4/3 - 34 + (4/3)(34)}{1 + (4/3)(34)}\right]$$

$$= \tan^{-1}\left[\frac{4/3 - 34 + (4/3)(34)}{1 + (4/3)(34)}\right]$$

$$= \tan^{-1}\left[\frac{4/3 - 34 + (4/3)(34)}{1 + (4/3)(34)}\right]$$

$$= \tan^{-1}\left[\frac{16 - 9121 + 1212}{1 + 1212}\right] = \tan^{-1}724$$

Let

$$\tan^{-1}724 = \theta_2 \Rightarrow \tan \theta_2 = 724$$

$$\Rightarrow \cos \theta_2 = 1/2425$$

$$\Rightarrow \theta_2 = \cos^{-1}1/2425$$

$$\text{Now } \cos[\sin^{-1}(45) - \tan^{-1}(34)] = \cos[\cos^{-1}1/2425] = 1/2425$$

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Ex 4.4

Question 1.

Find the principal value of

(i) $\sec^{-1}(2\sqrt{3})$

(ii) $\cot^{-1}(\sqrt{3})$

(iii) $\operatorname{cosec}^{-1}(-\sqrt{2})$

Solution:

(i) Let $\sec^{-1}(2\sqrt{3}) = \theta$

$$\Rightarrow \sec \theta = 2\sqrt{3}$$

$$\Rightarrow \cos \theta = \frac{1}{2\sqrt{3}} = \cos \frac{\pi}{6}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

(ii) Let $\cot^{-1}(\sqrt{3}) = \theta$

$$\Rightarrow \cot \theta = \sqrt{3}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{3}} = \tan \frac{\pi}{6}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

(iii) Let $\operatorname{cosec}^{-1}(-\sqrt{2}) = \theta$

$$\Rightarrow \operatorname{cosec} \theta = -\sqrt{2}$$

$$\Rightarrow \sin \theta = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = -\frac{\pi}{4}$$

Question 2.

Find the value of

(i) $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$

(ii) $\sin^{-1}(-1) + \cos^{-1}(1/2) + \cot^{-1}(2)$

(iii) $\cot^{-1}(1) + \sin^{-1}(-\sqrt{3}/2) - \sec^{-1}(-\sqrt{2})$

Solution:

(i) Let $\tan^{-1}\sqrt{3} = \alpha$ and $\sec^{-1}(-2) = \theta \Rightarrow \sec \theta = -2 \Rightarrow \cos \theta = -1/2 \Rightarrow \theta = \pi - \pi/3 = 2\pi/3$.
 $\therefore \tan^{-1}(\sqrt{3}) - \sec^{-1}(-2) = \alpha - (2\pi/3) = \pi/3 - (2\pi/3) = -\pi/3$

(ii) $\sin^{-1}(-1) = -\pi/2$ and $\cos^{-1}(1/2) = \pi/3$.
 $\therefore \sin^{-1}(-1) + \cos^{-1}(1/2) + \cot^{-1}(2) = -\pi/2 + \pi/3 + \cot^{-1}(2) = -\pi/6 + \cot^{-1}(2)$

(iii)

$$\cot^{-1}(1) = \pi/4, \sin^{-1}(-\sqrt{3}/2) = -\pi/3, \sec^{-1}(-\sqrt{2}) = \cos^{-1}(-1/\sqrt{2}) = \pi - \pi/4 = 3\pi/4$$

$$\text{So } \cot^{-1}(1) + \sin^{-1}(-\sqrt{3}/2) - \sec^{-1}(-\sqrt{2}) = \pi/4 - \pi/3 - 3\pi/4 = -5\pi/6$$

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Ex4.5

Question 1.

Find the value, if it exists. If not, give the reason for

non-existence.

(i) $\sin^{-1}(\cos \pi)$

(ii) $\tan^{-1}(\sin(-5\pi/2))$

(iii) $\sin^{-1}[\sin 5]$

Solution:

(i) $\sin^{-1}(\cos \pi) = \sin^{-1}(-1) = -\pi/2$

(ii) $\tan^{-1}(\sin(-5\pi/2)) = \tan^{-1}(-\sin \pi/2) = \tan^{-1}(-1) = -\pi/4$

(iii) $\sin^{-1}(\sin 5) = \sin^{-1}[\sin(5-2\pi)] = 5-2\pi$

Question 2.

Find the value of the expression in terms of x , with the help of a reference triangle.

(i) $\sin(\cos^{-1}(1-x))$

(ii) $\cos(\tan^{-1}(3x-1))$

(iii) $\tan(\sin^{-1}(x+1/2))$

Solution:

(i) Let $\cos^{-1}(1-x) = \theta$

$$1-x = \cos \theta$$

We know $\sin^2 \theta = 1 - \cos^2 \theta$

$$\Rightarrow \sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - (1-x)^2} = \sqrt{1 - (1+x^2-2x)} = \sqrt{2x-x^2} \Rightarrow \sin[\cos^{-1}(1-x)] = \sin \theta = \sqrt{2x-x^2}$$

(ii) Let $\tan^{-1}(3x-1) = \theta \Rightarrow 3x-1 = \tan \theta$

$$\text{So } \tan^2 \theta = (3x-1)^2 = 9x^2 - 6x + 1$$

$$\text{We know } 1 + \tan^2 \theta = \sec^2 \theta = \frac{1}{\cos^2 \theta} \Rightarrow \frac{1}{\cos^2 \theta} = 9x^2 - 6x + 1 + 1 = 9x^2 - 6x + 2$$

$$\text{So } \cos^2 \theta = \frac{1}{9x^2 - 6x + 2} \Rightarrow \cos \theta = \frac{1}{\sqrt{9x^2 - 6x + 2}}$$

$$\therefore \cos[\tan^{-1}(3x-1)] = \frac{1}{\sqrt{9x^2 - 6x + 2}}$$

(iii) Let $\sin^{-1}(x+1/2) = \theta \Rightarrow x+1/2 = \sin \theta \Rightarrow 2x+1 = 2\sin \theta \Rightarrow \sin \theta = \frac{2x+1}{2}$
 $\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \left(\frac{2x+1}{2}\right)^2} = \sqrt{1 - \frac{(4x^2+4x+1)}{4}} = \frac{1}{2}\sqrt{3-4x-4x^2}$
 $\therefore \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{2x+1}{\sqrt{3-4x-4x^2}}$

Question 3.

Find the value of

(i) $\sin^{-1}(\cos(\sin^{-1}(\sqrt{3}/2)))$

(ii) $\cot(\sin^{-1} 3/5 + \sin^{-1} 4/5)$

(iii) $\tan(\sin^{-1} 3/5 + \cot^{-1} 3/2)$

Solution:

(i) $\sin^{-1}(\sqrt{3}/2) = \pi/3$ and $\cos(\pi/3) = 1/2$ and so

$$\sin^{-1}[\cos\{\sin^{-1}(\sqrt{3}/2)\}] = \sin^{-1}(1/2) = \pi/6$$

(ii) Let $\sin^{-1}45 = \theta \Rightarrow \sin \theta = 45 \Rightarrow \cos \theta = 35 \Rightarrow \theta = \cos^{-1}35 \therefore \cot[\sin^{-1}35 + \sin^{-1}45] = \cot[\sin^{-1}35 + \cos^{-1}35] = \cot(\pi/2) = 0$

(iii) Let $\sin^{-1}35 = \theta$

$\Rightarrow \sin \theta = 35 \Rightarrow \tan \theta = 34$ so $\theta = \tan^{-1}34$ so $\theta = \tan^{-1}34$ Now let

$\cot^{-1}32 = \theta \Rightarrow \cot \theta = 32 \Rightarrow \tan \theta = 23 \Rightarrow \theta = \tan^{-1}23$ Now

$\sin^{-1}35 + \cot^{-1}32 = \tan^{-1}34 + \tan^{-1}23 = \tan^{-1}\left(\frac{34+231-34 \times 23}{9+8121-612}\right) = \tan^{-1}\left(\frac{176}{176}\right) = \tan^{-1}(1) = \pi/4$ so $\tan[\tan^{-1}(1)] = 1$

Question 4.

Prove that

(i) $\tan^{-1}211 + \tan^{-1}724 = \tan^{-1}12$

(ii) $\sin^{-1}35 - \cos^{-1}1213 = \sin^{-1}1665$

Solution:

(i)

$$\tan^{-1}211 + \tan^{-1}724 = \tan^{-1}\left[\frac{211+7241-211 \times 724}{1+211 \times 724}\right]$$

$$= \tan^{-1}\left[\frac{935-152764}{1+152764}\right] = \tan^{-1}\left[\frac{-151829}{152765}\right] = \tan^{-1}(-12)$$

$$= \tan^{-1}(-12) = -\tan^{-1}(12)$$

(ii) Let $\cos^{-1}1213 = \theta \Rightarrow \cos \theta = 1213$

so

$$\sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - 144169} = \sqrt{25169} = 513$$

$$\Rightarrow \theta = \sin^{-1}513$$

Now to find $\sin^{-1}35 - \sin^{-1}513$

We know $\sin^{-1}x - \sin^{-1}y = \sin^{-1}[x\sqrt{1-y^2} - y\sqrt{1-x^2}]$

$$\text{LHS: } \sin^{-1}35 - \sin^{-1}513 = \sin^{-1}[35\sqrt{1-25169} - 513\sqrt{1-925}] = \sin^{-1}[35 \times 1213 - 513 \times 45] = \sin^{-1}(3665 - 2065) = \sin^{-1}1665 = \text{RHS.}$$

Question 5.

Prove that $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \tan^{-1}[x+y+z - xyz1 - xy - yz - zx]$

Solution:

$$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y1-xy}{1+xy}\right) = \tan^{-1}(A) \text{ Here } A = \frac{x+y1-xy}{1+xy}$$

So LHS: $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \tan^{-1}(A) + \tan^{-1}z$

$$\tan^{-1}\left(\frac{A+z1-Az}{1+Az}\right) = \tan^{-1}\left[\frac{\left(\frac{x+y1-xy}{1+xy}\right) + z1 - \left(\frac{x+y1-xy}{1+xy}\right)z}{1 + \left(\frac{x+y1-xy}{1+xy}\right)z}\right] = \tan^{-1}\left[\frac{x+y+z - xyz1 - xy - yz - zx}{1 - xy - yz - zx}\right] = \text{RHS}$$

Question 6.

If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, show that $x+y+z=xyz$.

Solution:

$$\text{Given } \tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi \Rightarrow \tan^{-1}(x+y+z - xyz1 - xy - yz - zx) = \pi \Rightarrow x+y+z - xyz1 - xy - yz - zx = \tan \pi = 0 \Rightarrow x+y+z - xy - yz - zx = 0 \Rightarrow x+y+z = xyz$$

$$z=0 \Rightarrow x+y+z=xyz$$

Question 7.

Prove that

$$\tan^{-1}x + \tan^{-1}2x + \tan^{-1}3x = \tan^{-1}3x - x^3 - 3x^2, |x| < 1/\sqrt{3}$$

Solution:

$$\text{LHS: } \tan^{-1}x + \tan^{-1}2x + \tan^{-1}3x = \tan^{-1}\left[\frac{x + 2x + 3x - x(2x + 3x)}{1 - x(2x + 3x)}\right]$$

$$\left[\frac{x + 2x + 3x - x(2x + 3x)}{1 - x(2x + 3x)}\right]$$

$$= \tan^{-1}\left[\frac{x + 2x + 3x - x(2x + 3x)}{1 - x(2x + 3x)}\right]$$

$$\left[\frac{x + 2x + 3x - x(2x + 3x)}{1 - x(2x + 3x)}\right]$$

$$= \tan^{-1}\left[\frac{x + 2x + 3x - x(2x + 3x)}{1 - x(2x + 3x)}\right] = \tan^{-1}3x - x^3 - 3x^2 = \text{RHS}$$

Question 8.

Simplify: $\tan^{-1}xy - \tan^{-1}x - yx + y$

Solution:

$$\tan^{-1}xy - \tan^{-1}x - yx + y = \tan^{-1}\left[\frac{xy - x - yx + y}{1 + (xy)(x - yx + y)}\right]$$

$$\left[\frac{xy - x - yx + y}{1 + (xy)(x - yx + y)}\right]$$

$$= \tan^{-1}\left[\frac{xy - x - yx + y}{1 + (xy)(x - yx + y)}\right]$$

$$\left[\frac{xy - x - yx + y}{1 + (xy)(x - yx + y)}\right]$$

$$= \tan^{-1}\left[\frac{xy - x - yx + y}{1 + (xy)(x - yx + y)}\right]$$

$$\left[\frac{xy - x - yx + y}{1 + (xy)(x - yx + y)}\right]$$

$$= \tan^{-1}(x^2 + y^2xy + y^2 + x^2 - xy) = \tan^{-1}(x^2 + y^2x^2 + y^2) = \tan^{-1}(1) = \pi/4$$

Question 9.

Find the value of

$$(i) \sin^{-1}5x + \sin^{-1}12x = \pi/2$$

$$(ii) 2\tan^{-1}x = \cos^{-1}\frac{1-a^2}{1+a^2} - \cos^{-1}\frac{1-b^2}{1+b^2}, a > 0, b > 0$$

$$(iii) 2\tan^{-1}(\cos x) = \tan^{-1}(2\operatorname{cosec} x)$$

$$(iv) \cot^{-1}x - \cot^{-1}(x+2) = \pi/12, x > 0$$

Solution:

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(i) Given $\sin^{-1}5x + \sin^{-1}12x = \pi/2 \Rightarrow \sin^{-1}5x = \pi/2 - \sin^{-1}12x = \cos^{-1}12x$ ($\because \sin^{-1}x + \cos^{-1}x = \pi/2$) so now $\sin^{-1}5x = \cos^{-1}12x$ Let $\sin^{-1}5x = \theta$ Let

$\cos^{-1}12x = \theta \Rightarrow \sin\theta = 5x \Rightarrow 12x = \cos\theta$ But we know

$$\sin^2\theta + \cos^2\theta = 1 \Rightarrow (5x)^2 + (12x)^2 = 1 \text{ i.e. } 25x^2 + 144x^2 = 1 \Rightarrow x^2 = 1/169 \text{ a } x = 1/13$$

(ii)

$$\text{Let } 2\tan^{-1}x = \cos^{-1}\frac{1-a^2}{1+a^2} - \cos^{-1}\frac{1-b^2}{1+b^2}$$

Question 1.

The value of $\sin^{-1}(\cos x), 0 \leq x \leq \pi$ is

- (a) $\pi - x$
- (b) $x - \pi/2$
- (c) $\pi/2 - x$
- (d) $\pi - x$

Solution:

- (c) $\pi/2 - x$

Hint:

We know $\cos x = \sin(\pi/2 - x)$

So $\sin^{-1}(\cos x) = \sin^{-1}[\sin(\pi/2 - x)] = \pi/2 - x$

Question 2.

If $\sin^{-1}x + \sin^{-1}y = 2\pi/3$; then $\cos^{-1}x + \cos^{-1}y$ is equal to

- (a) $2\pi/3$
- (b) $\pi/3$
- (c) $\pi/6$
- (d) π

Solution:

- (b) $\pi/3$

Hint:

We know $\sin^{-1}x + \cos^{-1}x = \pi/2$

and $\sin^{-1}y + \cos^{-1}y = \pi/2$

Adding them

so $(\sin^{-1}x + \sin^{-1}y) + (\cos^{-1}x + \cos^{-1}y) = \pi$

i.e. $2\pi/3 + (\cos^{-1}x + \cos^{-1}y) = \pi$

$\Rightarrow \cos^{-1}x + \cos^{-1}y = \pi - 2\pi/3 = \pi/3$

Question 3.

$\sin^{-1}3/5 - \cos^{-1}4/5 + \sec^{-1}5/3 - \csc^{-1}5/2 =$

- (a) 2π
- (b) π
- (c) 0
- (d) $\tan^{-1}2/5$

Solution:

- (c) 0

Hint:

We know $\sin^{-1}35 + \cos^{-1}35 = \pi/2$

and $\cos^{-1}1213 + \sin^{-1}1213 = \pi/2$

so

$$\pi/2 - \pi/2 = 0$$

Question 4.

If $\sin^{-1}x = 2\sin^{-1}\alpha$ has a solution, then

(a) $|\alpha| \leq 1/\sqrt{2}$

(b) $|\alpha| \geq 1/\sqrt{2}$

(c) $|\alpha| < 1/\sqrt{2}$

(d) $|\alpha| > 1/\sqrt{2}$

Solution:

(a) $|\alpha| \leq 1/\sqrt{2}$

Hint:

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$\sin^{-1}x = 2\sin^{-1}\alpha$ Range of $\sin^{-1}x$ is $[-\pi/2, \pi/2]$ i.e.

$$-\pi/2 \leq \sin^{-1}x \leq \pi/2 \Rightarrow -\pi/2 \leq 2\sin^{-1}\alpha \leq \pi/2 \Rightarrow -\pi/4 \leq \sin^{-1}\alpha \leq \pi/4 \Rightarrow \sin^{-1}(-\pi/4) \leq \alpha \leq \sin(\pi/4) \Rightarrow -1/\sqrt{2} \leq \alpha \leq 1/\sqrt{2} \Rightarrow |\alpha| \leq 1/\sqrt{2}$$

Question 5.

$\sin^{-1}(\cos x) = \pi/2 - x$ is valid for

(a) $-\pi \leq x \leq 0$

(b) $0 \leq x \leq \pi$

(c) $-\pi/2 \leq x \leq \pi/2$

(d) $-\pi/4 \leq x \leq 3\pi/4$

Solution:

(b) $0 \leq x \leq \pi$

Question 6.

If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = 3\pi/2$, the value of $x^{2017} + y^{2018} + z^{2019} - 9x^{101} + y^{101} + z^{101}$ is

(a) 0

(b) 1

(c) 2

(d) 3

Solution:

(a) 0

Hint:

The maximum value of $\sin^{-1}x$ is $\frac{\pi}{2}$ and $\sin^{-1}1 = \frac{\pi}{2}$

Here it is given that

$$\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = 3\frac{\pi}{2} \Rightarrow x = y = z = 1 \text{ and}$$

$$\text{so } 1 + 1 + 1 - 9 = 1 + 1 + 1 - 3 = 0$$

Question 7.

If $\cot^{-1}x = 2\pi/5$ for some $x \in \mathbb{R}$, the value of $\tan^{-1}x$ is

(a) $-\pi/10$

(b) $\pi/5$

(c) $\pi/10$

(d) $-\pi/5$

Solution:

(c) $\pi/10$

Hint:

$$\text{Given } \cot^{-1}x = 2\pi/5 \Rightarrow x = \cot 2\pi/5$$

$$\text{We know } \tan^{-1}x + \cot^{-1}x = \pi/2$$

SO

$$\tan^{-1}x = \pi/2 - 2\pi/5 = \pi/10$$

Question 8.

The domain of the function defined by $f(x) = \sin^{-1}\sqrt{x-1}$ is

(a) $[1, 2]$

(b) $[-1, 1]$

(c) $[0, 1]$

(d) $[-1, 0]$

Solution:

(a) $[1, 2]$

Hint:

The domain for $\sin^{-1}x$ is $[0, 1]$

$$\text{So } \sqrt{x-1} = 0 \Rightarrow x-1 = 0 \Rightarrow x = 1$$

$$\sqrt{x-1} = 1 \Rightarrow x-1 = 1 \Rightarrow x = 2$$

$\therefore$ The domain is $[1, 2]$

Question 9.

If $x = 15$, the value of $\cos(\cos^{-1}x + 2\sin^{-1}x)$ is

(a) $-\sqrt{2/425}$

(b) $\sqrt{2/425}$

(c) 15

(d) -15

Solution:

(d) -15

Hint:

Now $\cos[\cos^{-1}15 + 2\sin^{-1}15] = \cos[\cos^{-1}15 + \sin^{-1}15 + \sin^{-1}15] = \cos[\pi/2 + \sin^{-1}15] = \cos[\pi/2 + \pi/2 - \cos^{-1}15] = \cos[\pi - \cos^{-1}15] = \cos\cos^{-1}(-15) = -15$

Question 10.

$\tan^{-1}(14) + \tan^{-1}(29)$ is equal to

(a) $12\cos^{-1}(35)$

(b) $12\sin^{-1}(35)$

(c) $12\tan^{-1}(35)$

(d) $\tan^{-1}(12)$

Solution:

(d) $\tan^{-1}12$

Hint:

$\tan^{-1}14 + \tan^{-1}29 = \tan^{-1}\left(\frac{14+29}{1-14 \times 29}\right) = \tan^{-1}\left(\frac{43}{-406}\right) = \tan^{-1}\left(\frac{1}{-9.44}\right) = \tan^{-1}(-0.106) = \tan^{-1}(-12)$

Question 11.

If the function $f(x) = \sin^{-1}(x^2 - 3)$, then x belongs to

(a) $[1, -1]$

(b) $[\sqrt{2}, 2]$

(c) $[-2, -\sqrt{2}] \cup [\sqrt{2}, 2]$

(d) $[-2, -\sqrt{2}] \cap [\sqrt{2}, 2]$

Solution:

(c) $[-2, -\sqrt{2}] \cup [\sqrt{2}, 2]$

Hint:

$f(x) = \sin^{-1}(x^2 - 3)$

Domain of $\sin^{-1}(x)$ is $[-1, 1]$

$\Rightarrow -1 \leq x^2 - 3 \leq 1 \Rightarrow 2 \leq x^2 \leq 4 \Rightarrow \sqrt{2} \leq |x| \leq 2 \Rightarrow x \in [-2, -\sqrt{2}] \cup [\sqrt{2}, 2]$

Question 12.

If $\cot^{-1}2$ and $\cot^{-1}3$ are two angles of a triangle, then the third angle is

- (a) $\pi/4$
- (b) $3\pi/4$
- (c) $\pi/6$
- (d) $\pi/3$

Solution:

- (b) $3\pi/4$

Hint:

Let $\cot^{-1}2 = \theta_1 \Rightarrow \cot \theta_1 = 2$ so $\tan \theta_1 = 1/2$

Let $\cot^{-1}3 = \theta_2 \Rightarrow \cot \theta_2 = 3$ so $\tan \theta_2 = 1/3$

$\tan(\theta_1 + \theta_2) = \frac{\tan \theta_1 + \tan \theta_2}{1 - \tan \theta_1 \tan \theta_2} = \frac{1/2 + 1/3}{1 - 1/2 \times 1/3} = \frac{5/6}{5/6} = 1 \Rightarrow \theta_1 + \theta_2 = \pi/4$

Let θ_3 be the third angle

So

$\theta_1 + \theta_2 + \theta_3 = 180^\circ = \pi \Rightarrow \theta_3 = \pi - \pi/4 = 3\pi/4$

Question 13.

$\sin^{-1}(\tan \pi/4) - \sin^{-1}(\sqrt{3}x) = \pi/6$. Then x is a root of the equation

- (a) $x^2 - x - 6 = 0$
- (b) $x^2 - x - 12 = 0$
- (c) $x^2 + x - 12 = 0$
- (d) $x^2 + x - 6 = 0$

Solution:

- (b) $x^2 - x - 12 = 0$

Hint:

$\sin^{-1}(1) - \sin^{-1}(\sqrt{3}x) = \pi/6 \Rightarrow \pi/2 - \sin^{-1}(\sqrt{3}x) = \pi/6 \Rightarrow \sin^{-1}(\sqrt{3}x) = \pi/2 - \pi/6 = \pi/3 \Rightarrow \sqrt{3}x = \sin \pi/3 = \sqrt{3}/2 \Rightarrow x = 1/2$

Question 14.

$\sin^{-1}(2\cos^2 x - 1) + \cos^{-1}(1 - 2\sin^2 x) =$

- (a) $\pi/2$.
- (b) $\pi/3$
- (c) $\pi/4$
- (d) $\pi/6$

Solution:

- (a) $\pi/2$

Hint:

$2\cos^2 x - 1 = \cos 2x$ and $1 - 2\sin^2 x = \cos 2x$.
 $\sin^{-1}(\cos 2x) + \cos^{-1}(\cos 2x) = \pi/2$ ($\because \sin^{-1} x + \cos^{-1} x = \pi/2$)

$$+\cos^{-1}x=\pi/2)$$

Question 15.

If $\cot^{-1}(\sqrt{\sin\alpha})+\tan^{-1}(\sqrt{\sin\alpha})=u$, then $\cos 2u$ is equal to

(a) $\tan 2\alpha$

(b) 0

(c) -1

(d) $\tan 2\alpha$

Solution:

(c) -1

Hint:

$$\cot^{-1}x+\tan^{-1}x=\pi/2 \Rightarrow u=\pi/2 \text{ so } 2u=\pi \therefore \cos 2u=\cos \pi=-1$$

Question 16.

If $|x|\leq 1$, then $2\tan^{-1}x-\sin^{-1}2x$ is equal to

(a) $\tan^{-1}x$

(b) $\sin^{-1}x$

(c) 0

(d) π

Solution:

(c) 0

Hint:

$$\text{Let } x=\tan\theta \text{ so } 2x+\sin 2\theta=\sin 2\theta.$$

$$\text{Now } 2\tan^{-1}(\tan\theta)-\sin^{-1}(\sin 2\theta)=2\theta-2\theta=0$$

Question 17.

The equation $\tan^{-1}x-\cot^{-1}x=\tan^{-1}(1/\sqrt{3})$ has

(a) no solution

(b) unique solution

(c) two solutions

(d) infinite number of solutions

Solution:

(b) unique solution

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Hint:

$$\tan^{-1}x-\cot^{-1}x=\tan^{-1}1/\sqrt{3} \Rightarrow \tan^{-1}x-(\pi/2-\tan^{-1}x)=\pi/6$$

$$2\tan^{-1}x=\pi/2+\pi/6 \Rightarrow 2\tan^{-1}x=2\pi/3 \Rightarrow x=\tan \pi/3=\sqrt{3}$$

$\Rightarrow$ It has a unique Solution

Question 18.

If $\sin^{-1}x+\cot^{-1}(1/2)=\pi/2$, then x is equal to

(a) 12

(b) $1/\sqrt{5}$

(c) $2/\sqrt{5}$

(d) $\sqrt{32}$

Solution:

(b) $1\sqrt{5}$

Hint:

$$\sin^{-1}x + \cot^{-1}(12) = \pi/2 \Rightarrow \sin^{-1}x = \pi/2 - \cot^{-1}12 = \tan^{-1}12 \Rightarrow \sin^{-1}x = \sin^{-1}\left(\frac{12}{\sqrt{1+(12)^2}}\right)$$

|

|

$$\left(\frac{12}{\sqrt{1+(12)^2}}\right)$$

|

|

$$\Rightarrow \sin^{-1}x = \sin^{-1}\left(\frac{12}{\sqrt{145}}\right) = 1\sqrt{5} \Rightarrow x = 1\sqrt{5}$$

Question 19.

If $\sin^{-1}x + \csc^{-1}5 = \pi/2$, then the value of x is

(a) 4

(b) 5

(c) 2

(d) 3

Solution:

(d) 3

Hint:

$$\begin{aligned} \sin^{-1}x + \csc^{-1}5 &= \pi/2 \Rightarrow \sin^{-1}x = \pi/2 - \csc^{-1}5 = \sec^{-1}5 \Rightarrow \sin^{-1}x = \sec^{-1}5 \\ &= \cos^{-1}4 \text{ Let } \sin^{-1}x = \theta \Rightarrow \sin\theta = x \text{ so} \\ \cos\theta &= 4/5 \Rightarrow \theta = \cos^{-1}4/5 = \cos^{-1}4/5 \text{ (given)} \\ \Rightarrow \sin\theta &= 3/5 \Rightarrow x = 3/5 \end{aligned}$$

Question 20.

$\sin(\tan^{-1}x)$, $|x| < 1$ is equal to

(a) $x\sqrt{1-x^2}$

(b) $1\sqrt{1-x^2}$

(c) $1\sqrt{1+x^2}$

(d) $x\sqrt{1+x^2}$

Solution:

(d) $x\sqrt{1+x^2}$

Hint:

$$\sin(\tan^{-1}x) = \sin(\sin^{-1}x\sqrt{1+x^2}) = x\sqrt{1+x^2} = x\sqrt{1+x^2}$$

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